

SERVQUAL

SERVQUAL is a research tool used to measure customer perception of service quality. It compares what customers expected from a service to how they think the service actually performed. The tool was developed in the United States in the mid-1980s by researchers A. Parasuraman, Valarie Zeithaml, and Leonard L. Berry.^[1]

SERVQUAL measures service quality across five areas: reliability, assurance, tangibles, empathy, and responsiveness. It has been used in both research and in applied settings (e.g., healthcare, banking, education, and libraries), sometimes with modifications for specific contexts.^{[1][2]} While SERVQUAL is one of the most commonly used methods for measuring service quality, it has also been criticized by researchers for how it defines expectations and for how well its five dimensions apply across different industries and cultures.

Overview

The SERVQUAL questionnaire consists of matched pairs of items, 22 expectation items and 22 perceptions items, organized into five dimensions which align with the consumer's mental map of service quality dimensions. ^[2] The questionnaire may be administered as a paper survey, web survey or in a face-to-face interview. Known studies have published scores for validity and reliability in small to large sample sizes.^{[3][4]} In practice, researchers often add supplementary items, such as respondent demographics, prior experience with the brand or service category, and behavioral intentions (e.g., intention to revisit or repurchase, loyalty intentions, and propensity to provide word-of-mouth referrals). As a result, the final questionnaire may include more than 60 items, although the original 22 SERVQUAL items remain unchanged. The face-to-face interview format may take up to one hour per respondent to administer, whereas print or web-based versions typically require less time.

Summary of SERVQUAL items ^[5]

Dimension	No. of Items in Questionnaire	Definition
Reliability	5	The ability to perform the promised service dependably and accurately
Assurance	4	The knowledge and courtesy of employees and their ability to convey trust and confidence
Tangibles	4	The appearance of physical facilities, equipment, personnel and communication materials
Empathy	5	The provision of caring, individualized attention to customer
Responsiveness	4	The willingness to help customers and to provide prompt service

In their application of SERVQUAL, many researchers make minor modifications to the instrument as necessary for context-specific applications. Some researchers label their revised instruments with new titles such as LibQUAL+ (libraries), EDUQUAL (educational context),^[6] HEALTHQUAL (hospital

context)^[7] and ARTSQUAL (art museum).^[8]

Examples of matched pairs of items in the SERVQUAL questionnaire^[9]

Dimension	Sample expectations item	Sample perceptions item
Reliability	When excellent telephone companies promise to do something by a certain time, they do so	XYZ company provides its services at the promised time
Assurance	The behaviour of employees in excellent banks will instill confidence in customers	The behaviour of employees in the XYZ bank instills confidence in you.
Tangibles	Excellent telephone companies will have modern looking equipment	XYZ company has modern looking equipment
Empathy	Excellent banks will have operating hours convenient to customers	XYZ bank has convenient operating hours
Responsiveness	Employees of excellent telephone companies will never be too busy to help a customer	XYZ employees are never too busy to help you

The SERVQUAL questionnaire has been described as "the most popular standardized questionnaire to measure service quality".^[10] It is widely used by service firms, most often in conjunction with other measures of service quality and customer satisfaction. The SERVQUAL instrument was developed as part of a broader conceptualization of how customers understand service quality. This conceptualization is known as the *model of service quality* or more popularly as the *gaps model*.

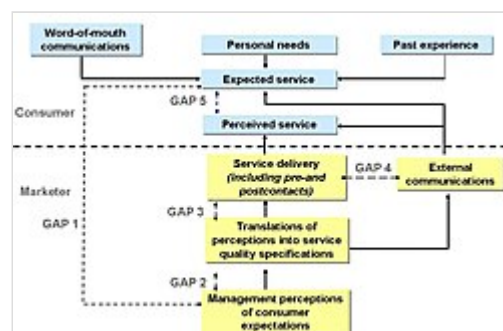
The model of service quality

The model of service quality, popularly known as the *gaps model* (<https://ca.indeed.com/career-advice/career-development/gap-model>), was developed by a group of American authors, A. Parasuraman, Valarie A. Zeithaml and Len Berry, in a systematic research program carried out in the 1980s.^[11] The model identifies the principal dimensions (or components) of service quality, proposes a scale for measuring service quality (SERVQUAL), and suggests possible causes of service quality problems. The model's developers originally identified ten dimensions of service quality, but after testing, some of the dimensions were found to be autocorrelated and the total number of dimensions was reduced to five, namely - reliability, assurance, tangibles, empathy and responsiveness. These five dimensions are thought to represent the dimensions of service quality across a range of industries and settings.^[12] The acronym RATER is often used among students of marketing.

Businesses use the SERVQUAL instrument (i.e. questionnaire) to measure potential service quality problems and the model of service quality to help diagnose possible causes of the problem. The model of service quality is built on the *expectancy–confirmation paradigm* which suggests that consumers perceive quality in terms of their perceptions of how well a given service delivery meets their expectations of that delivery.^[13] Thus, service quality can be conceptualized as a simple equation:

$$SQ = P - E$$

where;



A simplified model of service quality

SQ is service quality

P is the individual's perceptions of given service delivery

E is the individual's expectations of a given service delivery

When customer expectations are greater than their perceptions of received delivery, service quality is deemed low. When perceptions exceed expectations then service quality is high. The model of service quality identifies five gaps that may cause customers to experience poor service quality. In this model, gap 5 is the service quality gap and is the *only* gap that can be directly measured. In other words, the SERVQUAL instrument was specifically designed to capture gap 5. In contrast, Gaps 1–4 cannot be measured, but have diagnostic value.

Summary of Gaps with Diagnostic Indications ^[14]

Gap	Brief description	Probable Causes
Gap 1 The Knowledge Gap	Difference between the target market's expected service and management's perceptions of the target market's expected service	■ Insufficient marketing research ■ Inadequate upward communications ■ Too many layers of management
Gap 2 The standards Gap	Difference between management's perceptions of customer expectations and the translation into service procedures and specifications	■ Lack of management commitment to service quality ■ Employee perceptions of infeasibility ■ Inadequate goal setting ■ Inadequate task standardisation
Gap 3 The Delivery Gap	Difference between service quality specifications and the service actually delivered	■ Technical breakdowns or malfunctions ■ Role conflict/ ambiguity ■ Lack of perceived control ■ Poor employee-job fit ■ Poor technology- fit ■ Poor supervision or training
Gap 4 The Communications Gap	Difference between service delivery intentions and what is communicated to the customer	■ Lack of horizontal communications ■ Poor communication with advertising agency ■ Inadequate communications between sales and operations ■ Differences in policies and procedures across branches or divisions of an entity ■ Propensity to overpromise

Development of the instrument and model

The development of the model of service quality involved a systematic research undertaking which began in 1983, and after various refinements, resulted in the publication of the SERVQUAL instrument in 1988.^[15] The model's developers began with an exhaustive literature search to identify items that were believed to impact on perceived service quality. This initial search identified some 100 items which were used in the first rounds of consumer testing. Preliminary data analysis, using a data reduction technique known as factor analysis (also known as principal components analysis), revealed that these items loaded onto ten dimensions (or components) of service quality. The initial ten dimensions that were believed to represent service quality were:

1. **Competence** is the possession of the required skills and knowledge to perform the service. For example, there may be competence in the knowledge and skill of contact personnel, knowledge and skill of operational support personnel and research capabilities of the organization.
2. **Courtesy** is the consideration for the customer's property and a clean and neat appearance of contact personnel, manifesting as politeness, respect, and friendliness.
3. **Credibility** includes factors such as trustworthiness, belief and honesty. It involves having the customer's best interests at prime position. It may be influenced by company name, company reputation and the personal characteristics of the contact personnel.
4. **Security** enables the customer to feel free from danger, risk or doubt including physical safety, financial security and confidentiality.
5. **Access** is approachability and ease of contact. For example, convenient office operation hours and locations.
6. **Communication** means both informing customers in a language they are able to understand and also listening to the customers. A company may need to adjust its language for the varying needs of its customers. Information might include, for example, explanation of the service and its cost, the relationship between services and costs and assurances as to the way any problems are effectively managed.
7. **Knowing the customer** means making an effort to understand the customer's individual needs, providing individualized attention, recognizing the customer when they arrive and so on. This in turn helps to delight the customers by rising above their expectations.
8. **Tangibles** are the physical evidence of the service, for instance, the appearance of the physical facilities, tools and equipment used to provide the service; the appearance of personnel and communication materials and the presence of other customers in the service facility.
9. **Reliability** is the ability to perform the promised service in a dependable and accurate manner. The service is performed correctly on the first occasion, the accounting is correct, records are up to date and schedules are kept.
10. **Responsiveness** is the readiness and willingness of employees to help customers by providing prompt timely services, for example, mailing a transaction slip immediately or setting up appointments quickly.

Further testing suggested that some of the ten preliminary dimensions of service quality were closely related or autocorrelated. Thus the ten initial dimensions were reduced and the labels amended to accurately reflect the revised dimensions. By the early 1990s, the authors had refined the model to five factors which in testing, appear to be relatively stable and robust.

1. **Reliability:** the ability to perform the promised service dependably and accurately

2. **Assurance:** the knowledge and courtesy of employees and their ability to convey trust and confidence
3. **Tangibles:** the appearance of physical facilities, equipment, personnel and communication materials
4. **Empathy:** the provision of caring, individualized attention to customers
5. **Responsiveness:** the willingness to help customers and to provide prompt service

These are the five dimensions of service quality that form the basis of the individual items in the SERVQUAL research instrument (questionnaire). The acronym RATER is often used to help students of marketing remember the five dimensions of quality explicitly mentioned in the research instrument. It is these five dimensions that are believed to represent the consumer's mental checklist of service quality.

Nyeck, Morales, Ladhari, and Pons (2002) stated that the SERVQUAL measuring tool "appears to remain the most complete attempt to conceptualize and measure service quality" (p. 101). The SERVQUAL measuring tool has been used by many researchers across a wide range of service industries and contexts, such as healthcare, banking, financial services, and education (Nyeck, Morales, Ladhari, & Pons, 2002).^[16]

Criticisms of SERVQUAL and the model of service quality

Although the SERVQUAL instrument has been widely applied in a variety of industry and cross-cultural contexts, there are many criticisms of the approach. Francis Buttle published one of the most comprehensive criticisms of the model of service quality and the associated SERVQUAL instrument in 1996 in which both operational and theoretical concerns were identified.^[17] Some of the more important criticisms include:

Face validity: The model of service quality has its roots in the expectancy-disconfirmation paradigm that informs customer satisfaction.^[18] Many researchers have argued that the research instrument actually captures *satisfaction* rather than *service quality*.^[19] Other researchers have questioned the validity of conceptualising service quality as a gap.^[20]

Construct validity: The model's developers tested and retested the SERVQUAL scale for reliability and validity. However, at the same time, the model's developers recommended that applied use of the instrument should modify or adapt them for specific contexts. Any attempt to adapt or modify the scale will have implications for the validity of items with implications for the validity of the dimensions of reliability, assurance, tangibles, empathy and responsiveness.^[21]

Ambiguity of expectations construct: SERVQUAL is designed to be administered after respondents have experienced a service. They are therefore asked to *recall* their pre-experience expectations. However, recall is not always accurate and raises concerns about whether the research design accurately captures true pre-consumption expectations. In addition, studies show that expectations actually change over time. Consumers are continually modifying their expectations as they gain experience with a product category or brand.^[22]

Operational definition of the expectations construct: The way that expectations has been operationalised also represents a concern for theorists investigating the validity of the gaps model. The literature identifies different types of expectations.^[23] Of these, there is an argument that only *forecast expectations* are true expectations. Yet, the SERVQUAL

instrument appears to elicit *ideal expectations*.^[24] Note the wording in the questionnaire in the preceding figure which grounds respondents in their expectations of what *excellent* companies will do. Subtle use of words can elicit different types of expectations. Capturing true expectations is important because it has implications for service quality scores. When researchers elicit ideal expectations, overall service quality scores are likely to be lower, making it much more difficult for marketers to deliver on those expectations.^[25]

Questionnaire length: The matched pairs design of the questionnaire (total of 22 expectation items plus 22 perception items= 44 total items) makes for a very long questionnaire. If researchers add demographic and other behavioural items such as prior experience with product or category and the standard battery of demographics including: age, gender, occupation, educational attainment etc. then the average questionnaire will have around 60 items. In practical terms, this means that the questionnaire would take more than one hour per respondent to administer in a face-to-face interview. Lengthy questionnaires are known to induce *respondent fatigue* which may have potential implications for data reliability. In addition, lengthy questionnaires add to the time and cost involved in data collection and data analysis. Coding, collation and interpretation of data is very time consuming and in the case of lengthy questionnaires administered across large samples, the findings cannot be used to address urgent quality-related problems. In some cases, it may be necessary to carry out 'quick and dirty' research while waiting for the findings of studies with superior research design.

Administration of the questionnaire: Some analysts have pointed out that the SERVPERF instrument, developed by Cronin and Taylor,^{[26][27]} and which reduced the number of questionnaire items by half (22 perceptions items only), achieves results that correlate well with SERVQUAL, with no reduction in diagnostic power, improved data accuracy through reductions in respondent boredom and fatigue and savings in the form of reduced administration costs.

Dimensional instability: Many studies have reported that the five dimensions of service quality implicit in the model (reliability, assurance, tangibles, empathy and responsiveness) do not hold up when the research is replicated in different countries, different industries, in different market segments or even at different time periods.^{[28][29]} Some studies report that the SERVQUAL items do not always load onto the same factors. In some empirical research, the items load onto fewer dimensions, while other studies report that the items load onto more than five dimensions of quality. In statistical terms, the robustness of the factor loadings is known as a model's *dimensional stability*. Across a wide range of empirical studies, the factors implicit in the SERVQUAL instrument have been shown to be unstable.^[30] Problems associated with the stability of the factor loadings may be attributed, at least in part, to the requirement that each new SERVQUAL investigation needed to make context-sensitive modifications to the instrument in order to accommodate the unique aspects of the focal service setting or problem. However, it has also been hypothesised that the dimensions of service quality represented by the SERVQUAL research instrument fail to capture the true dimensionality of the service quality construct and that there may not be a universal set of service quality dimensions that are relevant across all service industries.^[31]

In spite of these criticisms, the SERVQUAL instrument, or any one of its variants (i.e. modified forms), dominates current research into service quality.^[32] In a review of more than 40 articles that made use of SERVQUAL, a team of researchers found that "few researchers concern themselves with the validation of the measuring tool".^[16] SERVQUAL is not only widely covered as a subject of academic papers, but also widely used by industry practitioners.^[33]

See also

- [Customer satisfaction](#)
- [Customer satisfaction research](#)
- [Disconfirmed expectancy](#)
- [Quality management](#)
- [Service quality](#)
- [Services marketing](#)

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External links

- [SERVQUAL Instructions \(http://www.kinesis-cem.com/pdf/ServQual.pdf\)](http://www.kinesis-cem.com/pdf/ServQual.pdf) - Detailed instructions for administering the SERVQUAL questionnaire

Further reading

- Luis Filipe Lages & Joana Cosme Fernandes, 2005, "The SERPVAL scale: A multi-item instrument for measuring service personal values", *Journal of Business Research*, Vol.58, Issue 11, pp 1562–1572.
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